

ROI ET AGROMET, Thailand

WMO: 484040

Lat: 16.0750N

Lon: 103.6044E

Elev: 511

StdP: 14.43

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	57.4	60.1	48.3	51.1	62.7	51.1	56.8	64.4	18.3	73.8	16.1	73.8	4.0	40	0.338

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	17.5	99.7	78.2	97.6	78.4	95.5	78.5	82.3	90.2	81.7	89.6	81.1	88.9	5.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.6	162.1	85.5	79.9	158.2	85.0	79.2	154.7	84.6	46.7	90.3	46.0	90.0	45.4	89.3	93.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	15.1	13.4	11.8	DB	52.3	101.8	2.3	2.4	50.7	103.5	49.4	104.9	48.1	106.2	46.5	107.9
(5)				WB	49.5	85.0	2.7	2.9	47.6	87.1	46.0	88.7	44.5	90.3	42.5	92.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.4	74.5	78.4	83.1	86.1	85.4	84.9	83.3	82.8	82.3	81.3	79.2	74.9
	DBStd	5.27	4.77	5.48	5.25	4.18	2.90	2.39	2.75	2.10	2.08	2.30	3.54	4.77
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	4	2	1	0	0	0	0	0	0	0	0	0	1
	CDD50	11446	760	795	1027	1083	1098	1047	1033	1017	970	970	875	771
	CDD65	5974	297	375	562	633	633	597	568	552	520	505	425	307
	CDH74	65152	2529	3841	6596	7967	7718	7317	6390	6061	5431	4935	3818	2549
	CDH80	27636	898	1768	3359	4246	3693	3254	2460	2127	1758	1713	1449	911
Wind	WSAvg	4.0	4.7	3.9	3.9	3.4	3.3	4.3	4.4	4.0	2.8	3.5	4.7	5.4
Precipitation	PrecAvg	57.5	0.2	0.7	1.3	3.5	8.2	7.8	8.6	10.7	12.1	3.9	0.6	0.1
	PrecMax	73.5	1.6	3.9	5.3	7.1	19.0	11.7	16.0	18.0	22.9	8.8	2.5	1.0
	PrecMin	41.1	0.0	0.0	0.0	0.4	2.2	3.5	3.8	1.9	4.8	0.7	0.0	0.0
	PrecStd	8.8	0.4	1.1	1.4	2.0	4.4	2.4	3.3	4.8	4.8	2.3	0.8	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.8	97.5	101.7	103.1	99.7	97.3	96.9	92.6	91.8	91.9	93.4	92.9
		MCWB	75.2	75.3	75.5	77.3	80.0	80.8	78.3	80.4	80.9	78.3	78.6	75.4
	2%	DB	90.0	95.1	98.9	100.4	97.5	94.9	93.5	90.9	90.4	90.2	91.1	90.5
		MCWB	74.4	75.3	76.2	79.0	80.0	80.0	79.5	80.3	80.4	78.0	77.7	74.3
	5%	DB	87.8	92.7	96.5	97.9	95.2	92.9	91.1	89.4	89.0	89.0	89.5	88.1
		MCWB	73.4	75.1	77.0	79.2	80.1	79.8	79.6	79.9	80.0	77.6	76.7	73.9
	10%	DB	85.4	90.0	93.9	95.2	92.8	91.1	89.0	88.0	87.6	87.7	87.6	85.3
		MCWB	71.8	74.8	77.1	78.7	80.1	79.7	79.6	79.7	79.8	77.4	75.6	72.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.0	79.3	81.0	82.8	83.1	82.9	81.9	82.5	82.8	81.1	80.6	77.7
		MCDB	88.4	90.6	93.2	94.0	91.7	91.6	89.0	88.3	87.9	86.7	89.5	89.0
	2%	WB	77.2	78.2	79.6	81.4	81.8	81.9	81.0	81.6	81.8	80.3	79.6	76.3
		MCDB	85.3	89.5	92.0	93.4	91.1	90.5	88.2	87.6	87.4	85.9	88.0	87.2
	5%	WB	75.5	77.1	78.5	80.5	81.1	81.0	80.6	80.9	81.1	79.6	78.4	75.0
		MCDB	83.3	88.4	91.3	92.7	90.8	89.4	87.8	86.9	86.5	85.1	86.1	85.2
	10%	WB	73.7	75.9	77.7	79.6	80.6	80.5	80.0	80.4	80.5	78.9	77.4	73.7
		MCDB	82.5	86.6	90.2	91.4	90.5	88.6	86.8	86.2	85.8	84.4	84.4	83.4
Mean Daily Temperature Range	5% DB	MDBR	21.1	21.5	19.5	17.5	14.6	12.2	10.8	10.3	10.3	12.7	16.4	19.8
		MCDBR	22.4	23.4	22.2	19.9	17.5	14.7	14.0	12.1	12.2	14.2	17.9	21.6
		MCWBR	8.4	7.5	6.1	5.7	5.0	4.2	4.2	4.4	4.8	5.3	6.1	7.6
	5% WB	MCDBR	18.7	19.8	19.5	18.1	15.6	13.3	12.0	11.3	11.5	12.7	16.0	19.8
		MCWBR	7.8	7.0	6.5	6.1	5.7	4.7	4.6	4.7	5.1	5.4	5.9	8.3
Clear-Sky Solar Irradiance	taub	0.476	0.593	0.661	0.619	0.513	0.469	0.459	0.463	0.506	0.525	0.471	0.465	
	taud	2.085	1.791	1.682	1.818	2.139	2.281	2.311	2.289	2.157	2.071	2.196	2.146	
	Ebn at Noon	257	232	221	231	253	262	265	266	256	247	257	256	
	Edh at Noon	49	69	80	70	50	43	42	43	49	52	44	46	
All-Sky Solar Radiation	RadAvg	1564	1650	1680	1812	1768	1654	1551	1509	1473	1599	1568	1512	
	RadStd	88	81	74	79	95	81	113	99	109	107	56	39	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	+0.49	N/A	N/A	N/A	+0.27	+0.23	N/A	N/A	N/A	N/A

Nomenclature: See separate page